Exp 10a – Types of Reactions

INTRODUCTION

There are a nearly infinite number of chemical reactions possible. Because of this, it is helpful to break them down into a few different classes of reactions. In this experiment, you will carry out and observe characteristic reaction(s) for each type.

BACKGROUND

At the introductory or general chemistry level, there are two basic schemes to categorize reactions- that help us understand and predict what will happen when two chemicals are mixed. Because they describe different aspects of the reaction, they are best understood together.

Scheme 1
What is happening with the atoms?

Double Displacement ("DD") ⟶
AB + CD → CB + AD

Single Displacement ("SD")
A + BC → B + AC

Combination
A + B → AB

Decomposition
AB → A + B

Scheme 2
What overall process is taking place?

Precipitation
A "DD" reaction that forms a solid product

Acid-Base/Neutralization
A "DD" reaction between an acid and base.

Gas Evolution
A "DD" reaction that forms a gaseous product.

No Reaction
A "DD" reaction with only aqueous products.

Oxidation/Reduction (Redox)
A reaction where electrons are transferred.
(Often "SD", Combination, or Decomposition)

↳ Combustion
A Redox reaction with gaseous oxygen.

Precipitation Reactions

A precipitation reaction is a double displacement reaction where two aqueous solutions are mixed, and an ionic solid is formed in the product.

Generic Form:	Salt(aq)	+	Salt(aq)	→	New Salt(s)	+ New Salt(aq)
Example:	$2NaCl(aq)$	+	$Pb(NO_3)_2(aq)$	→	$PbCl_2(s)$	+ $2NaNO_3(aq)$

Acid-Base Reactions

An acid-base reaction (also called a neutralization reaction) is a double displacement reaction between an acid and a base where water (or another weak acid) is formed as one of the products.

Generic Form: Acid(aq) + Base(aq) → Salt(aq) + Water(l)

Example: $HNO_3(aq) + KOH(aq) \rightarrow KNO_3(aq) + H_2O(l)$

Gas Evolution Reactions

A gas evolution reaction is a double displacement reaction where a gas is formed as one of the products. Commonly the gas is formed when one of the aqueous products formed decomposes into water and a molecular gas. The common products that decompose are H_2CO_3, H_2SO_3, and NH_4OH.

Generic Form: A) Salt(aq) + Acid(aq) → New Acid(aq) + New Salt(aq)

B) New Acid(aq) → Gas(g) + Water(l)

Example: A) $Na_2SO_3(aq) + 2HCl(aq) \rightarrow H_2SO_3(aq) + 2NaCl(aq)$

B) $H_2SO_3(aq) \rightarrow SO_2(g) + H_2O(l)$

No Reaction

If there is no observable change when reactants are mixed, then the reaction should be classified as "no reaction". This can be seen in the chemical equation when a double displacement reaction forms only aqueous products.

Redox Reactions

Oxidation-Reduction (redox) reactions involve the transfer of electrons. Although identifying these reactions can involve further calculations, most can be identified simply by the presence of a neutral, uncombined element in either the reactants or products. (As in Single Displacement, Combination, and Decomposition reactions.)

Examples: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

$2Al(s) + 3Cl_2(g) \rightarrow 2AlCl_3(s)$

$Cu(s) + Pd(NO_3)_2(aq) \rightarrow Cu(NO_3)_2(aq) + Pd(s)$

* Note that if, as in the first example above, a gas is produced as a result of a redox reaction, it is best to classify the reaction as redox and not gas evolution.

Combustion Reactions

A combustion reaction is a special type of redox reaction, where a hydrocarbon (compound composed mainly of hydrogen and carbon) or a metal reacts with oxygen gas. Although these are a type of redox reaction, they are so common that they should be classified as their own type of reaction.

Generic Form:	Hydrocarbon(g, l, or s)	+	Oxygen(g)	→	Carbon Dioxide(g)	+ Water(g)
Example:	$C_2H_4(g)$	+	$3O_2(g)$	→	$2CO_2(g)$	+ $2H_2O(g)$
Generic Form:	Metal(s)	+	Oxygen(g)	→	oxide of metal(s)	
Example:	2K(s)	+	$O_2(g)$	→	$K_2O(s)$	

Molecular and Complete Ionic Equations

There are different formats for writing chemical equations. All the equations above are written as "molecular equations" where all products are written as intact compounds. However, in aqueous solution, strong electrolytes dissociate into separate ions. The complete ionic equations demonstrate this:

	Aqueous Strong Electrolytes	**Solids, Liquids, and Gases**		
Molecular Equation	$2NaNO_3(aq)$	$PbCl_2(s)$	$H_2O(l)$	$CO_2(g)$
	Written as separate ions:	Written as intact compounds:		
Complete Ionic Equation	$2Na^+(aq) + 2NO_3^-(aq)$	$PbCl_2(s)$	$H_2O(l)$	$CO_2(g)$

ADDITIONAL READING

- Chemical Reactions, Molecular, Complete Ionic and Net Ionic Equations

SAFETY EQUIPMENT REQUIRED FOR LAB

- Goggles
- Lab coat
- Nitrile gloves (provided)

SUPPLY LIST

- Test tubes
- Test Tube Rack

CLASSROOM SUPPLY AND REACTANT LIST

- HCl (6.0 M)
- NaOH (3.0 M)
- $BaCl_2$ (0.1 M)
- Na_3PO_4 (0.1 M)
- $CaCl_2$ (0.1 M)
- Na_2CO_3 (0.1 M)
- Na_2CO_3 (1 M)
- KNO_3 (0.1 M)
- $CuSO_4$ (0.1 M)
- Na_2CO_3 (solid)
- $KClO_3$ (solid)
- NH_4Cl (4.0 M)
- Zinc pellets
- Red litmus paper
- Transfer pipets
- Spatula
- Matches

PROCEDURE

General Notes

a. All reactions are to be carried out in test tubes. All quantities may be considered approximate.
b. It is necessary to observe the reaction mixture for at least a few minutes to determine the character of any precipitate (solid) that forms.
c. When 1 mL of a reagent is specified, you may use approximately 20 drops from a Pasteur/transfer pipet or just pour one finger width into the test tube. There is no need to measure it precisely.
d. When a reaction calls for 1 g of a reactant this is a pea-sized quantity. There is no need to mass it with precision.
e. Record your own observations; photocopies or hand copies of someone else's work are not acceptable.
f. The reagents are set up by station number around the lab room. You are free to run the reactions in any convenient order.
g. Dump all waste into the waste containers provided. In chemistry labs, *never dump any waste down the drain* unless specifically instructed to do so.

Station 1. Transfer about 3 mL of 6.0 M HCl to a test tube. Add a few zinc pellets to the tube. Record your observations. Is the gas that evolves CO_2 or some other non-flammable gas? Or is it H_2 gas which is very flammable when it combusts with the formation of water?

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

* Evaluate the flammability of the gas by inverting a test tube over the one containing the reaction mixture. Capture a sample of the gas and light it with a match. If the gas is hydrogen, the combustion will yield a pop sound.

Clean-up for Station 1:

Dump test tube into temporary waste beaker at the station. Use forceps to remove any large zinc pellets. Rinse with DI water and place on paper towel. These zinc pellets can be used by other students.

Station 2. In a test tube mix 1 mL of 0.1 M $BaCl_2$ with 1 mL of 0.1 M Na_2SO_4.

What happens when you mix the two reactants? What does the test tube look like after 3-5 minutes? As part of your observations look-up the solubility of sodium chloride and barium sulfate on the solubility table handout. Based on the reactants and the given product what is the formula for the insoluble product you observe in this experiment?

Station 3. In a test tube mix 1 mL of 0.1 M $BaCl_2$ with 1 mL of 0.1 M Na_3PO_4.
What happens when you mix the two reactants? What does the test tube look like after 3-5 minutes? As part of your observations, look up the solubility of sodium chloride and barium phosphate on the solubility table handout. Based on the reactants and the given product what is the formula for the insoluble product you observe in this experiment?

Station 4. In a test tube mix 2 mL of 1.0 M Na_2CO_3 with 1 mL of 6.0 M HCl.
If a gas evolves, what compound is responsible for it? What happens to the solution over time? Report the overall chemical reaction that accounts for the production of any gas you observe.

Station 5. Transfer about 1 g of $KClO_3$ to a dry test tube (a small scoop). Place a test tube clamp near the top of the test tube. In the fume hood, heat the test tube using a Bunsen burner (blue flame). Hold the test tube at an angle so that only the bottom of the test tube is in the flame. Continue heating after the $KClO_3$ melts and bubbles form. Heat until nothing appears to be happening. What does the test tube look like? Did you see anything come out of the neck of the test tube? Record your observations associated with this station.

Station 6. In a test tube mix 1 mL of 0.1 M $CaCl_2$ with 1 mL of 0.1 M Na_2CO_3.
What happens when you mix the two reactants? What does the test tube look like after 3-5 minutes? As part of your observations look-up the solubility of sodium chloride and calcium carbonate in the solubility table. Based on the reactants and the given product what is the formula of the insoluble product you observe in this experiment?

Station 7. In a test tube mix 1 mL of 6.0 M HCl with 2 mL of 3.0 M NaOH.
What happens when you mix the two reactants? Is there any temperature change? What kind of reaction is this? What are the identities of the products you expect from this experiment?

Station 8. In a test tube mix 1 mL of 0.1 M KNO_3 with 1 mL of 0.1 M $BaCl_2$.
What happens when you mix the two reactants? What does the test tube look like after 3-5 minutes? As part of your observations look-up the solubility of potassium chloride and barium nitrate on the solubility table. Does this reaction yield a product? If not, why?

Station 9. Transfer about 2 mL of 0.1 M $CuSO_4$ to a test tube. Add a few zinc pellets to the tube. Shake for approximately 1 minute. What happens when you mix the two reactants? What does the test tube look like after 3-5 minutes (both pellets and solution color)? As part of your observations, look up the activity of zinc and copper on the activity series table handout. Based on the reactants and the given products what is the identity of the solid product you observe in this experiment? What accounts for the blue color of the $CuSO_4$ solution?

Station 10. In a test tube mix 1 mL of 4.0 M of NH_4Cl with 1 mL of 3.0 M NaOH.
If a gas evolves, identify it. Hold a piece of moistened red litmus over the test tube without touching the liquid solution in the test tube. A color change from red to blue confirms ammonia as a product. Do you detect any odor coming from the test tube? A pungent smell is also a confirmation of ammonia (NH_3). What happens to the solution over time? Report the overall chemical reaction that accounts for the production of any gas you observe.

CLEAN-UP AND WASTE DISPOSAL

1. ☐ After checking your data with your instructor, dump all test tubes in the aqueous inorganic waste container.
    a. If large quantities of solids are lodged inside the test tube, use a water bottle filled with DI water to rinse the solids into the aqueous waste container.
2. ☐ Using a small test tube brush, wash the test tubes with soapy water. Rinse with DI water.
3. ☐ Place the clean wet test tubes upside down in the test-tube rack to drain (if available), or put into a bin designated for used test tubes.

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

DATE _____

PARTNER _____

Record your observations then complete and balance the chemical reaction equation. Identify each chemical reaction equation by type. Include all the data requested in the procedure. Make your own observations; don't simply copy someone else's.

Station 1. Zn(s) and HCl(aq)

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

__Zn(s) + ___HCl(aq) → _____ + _____

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 2. $BaCl_2(aq)$ and $Na_2SO_4(aq)$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$BaCl_2(aq)$ + ___$Na_2SO_4(aq)$ → ___$BaSO_4$(___) + _____

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 3. $BaCl_2(aq)$ and $Na_3PO_4(aq)$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$BaCl_2(aq)$ + ___$Na_3PO_4(aq)$ → _____ + ___NaCl(___)

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 4. $HCl(aq)$ and $Na_2CO_3(aq)$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$HCl(aq)$ + ___$Na_2CO_3(aq)$ → _____+_____ + ___NaCl(___)

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 5. $KClO_3$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

____$KClO_3$ (s) → _____+ _____

Balanced Molecular Reaction Equation

This station does not take place in aqueous solution so you will not write complete ionic or net ionic equations for this process.

Station 6. $CaCl_2$(aq) and Na_2CO_3(aq)

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$CaCl_2$(aq) + ___Na_2CO_3(aq) → _____ + ___NaCl(___)

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 7. HCl(aq) and NaOH(aq)

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___HCl(aq) + ___NaOH(aq) → _____ + ___NaCl(___)

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 8. KNO_3(aq) and $BaCl_2$(aq)

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___KNO_3(aq) + ___$BaCl_2$(aq) → _____ + ___KCl(___)

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 9. $Zn(s)$ and $CuSO_4(aq)$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$Zn(s)$ + ___$CuSO_4$ (aq) → _____ +_____

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

Station 10. $NH_4Cl(aq)$ and $NaOH(aq)$

Reaction Type, Scheme 1:_____ Scheme 2:_____

Observations:

___$NH_4Cl(aq)$ + ___$NaOH(aq)$ → _____ + _____ + ___$NaCl($___$)$

Balanced Molecular Reaction Equation

_____→_____

Complete Ionic Reaction Equation

_____→_____

Net Ionic Reaction Equation

QUESTIONS

1. Identify each of the following as a salt, acid, base, or hydrocarbon.

a. Ammonium hydroxide _____

b. CaC_2O_4 _____

c. $Ba(ClO_4)_2$ _____

d. $HC_2H_3O_2$ _____

e. C_2H_6O _____

f. $H_2S_{(aq)}$ _____

2. Balance the following chemical reaction equations.

a. ____$Na_2S(aq)$ + ___$CaCl_2(g)$ → ___$CaS(s)$ + ___$NaCl(aq)$

b. ____$C_6H_{18}O(g)$ + ___$O_2(g)$ → ___$CO_2(g)$ + _____

c. ____$C_4H_9OH(g)$ + ___$O_2(g)$ → ___$CO_2(g)$ + _____

d. ____$Ca(OH)_2(s)$ +___$H_3PO_4(aq)$ → _____+ ___$H_2O(l)$

e. ____$NaHCO_3(s)$ +___$H_2SO_4(aq)$ → _____(aq) + _____(g) + ___ $H_2O(l)$

f. ____$Cd(s)$ + ____ $H_3PO_4(aq)$ → _____(s) + ___$H_2(g)$